

Name: _____

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Class: _____

MyDesign Portfolio

Element E Initial Template - AT - 15 points

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

This section should describe how your proposed solution is supported by STEM (science, technology, engineering, mathematics, computer science, etc.) concepts.

1. Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.
2. Be sure to apply the applicable STEM concepts in context, directly to the specifics of your design. Review concepts learned in your other classes and tackle new concepts as appropriate. Include any relevant calculations.
3. Element C indicates that you should have at least three high-priority design requirements to receive a score of 3 (and four for a score of 4, and five for a 5). Your number of design requirements and your number of high-priority design requirements should be the same throughout your portfolio. You should use the same three to five design requirements here too, and apply **at least three of them** to STEM concepts to achieve a score of 5 on E1. Use as many copies of this sub-template as needed.

High-Priority Design Requirement #1 from Element C -

STEM Concepts Involved in the requirement: **Biomimicry (modeling designs after nature), Biocompatibility (non-toxic interaction with living tissue), and System-Level Thinking to minimize environmental impact**

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Describe how STEM concepts apply directly to the specifics of your design requirement: **Our design requirement focuses towards the safety of the environment with the outer layer material the design has, this concept has a strong connection to what we have been aiming for.**

High-Priority Design Requirement #2 from Element C -

STEM Concepts Involved in the requirement: **Durability, Material Resilience, and Structural Stability against elements like wind, water, and temperature fluctuation with main concepts being Science and Technology**

Describe how STEM concepts apply directly to the specifics of your design requirement: **This concept applies to the materialistic structure that we have focussed on. We want the design to be really durable since it's a big design, so there will be strong qualities for going about this project.**

High-Priority Design Requirement #3 from Element C -

STEM Concepts Involved in the requirement: **Optimizing root development, nutrient uptake, and water infiltration. Key principles include reducing compaction, enhancing porosity, and ensuring nutrient bioavailability**

Describe how STEM concepts apply directly to the specifics of your design requirement:



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This concept can be applied to the qualities that are inside of the design. We point towards successfully balancing, possibly increasing the rate of photosynthesis, ensuring a healthy environment for the living things.

High-Priority Design Requirement #4 from Element C (if you have one)-

STEM Concepts Involved in the requirement: **Optimizing photosynthesis in poor light, managing resource competition, and utilizing genetic adaptations**

Describe how STEM concepts apply directly to the specifics of your design requirement:
The concept connects to the ideas we've provided to make sure that any poor condition will never happen. One of our important ideas is a light for the days that are dark.

High-Priority Design Requirement #5 from Element C (if you have one) -

STEM Concepts Involved in the requirement: **Incorporating concepts such as geometry, physics, advanced engineering tools, and data modeling. The goal is to minimize human error and account for terrain variability (elevation changes) to calculate precise areas, volumes, and boundaries**

Describe how STEM concepts apply directly to the specifics of your design requirement:
This concept applies to the building process of the design. Through the steps and procedures of going about this project, we check to see if the objects that we would use are applicable against our constraints.



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Proposed Solution to STEM Principles and Practices (E2)

While E1 was focused on how STEM applies to the design requirements, E2 is focused on how STEM is used to support the design solution.

For E2, use STEM concepts to justify your design decisions. It's important that the STEM Principles and Practices that are discussed are actually appropriate and useful to the problem. Use as many copies of this sub-template as needed.

STEM Principles for Design Solution #1

STEM Concept: **Incorporate composting and organic matter to rejuvenate degraded, compacted soil. Choose native grasses and companion plants that are drought-resistant and suitable for the local ecosystem. Maintain living roots year-round to feed microbes and improve soil structure.**

Describe how STEM concept applies directly to the specifics of the first solution to your problem: **The greenhouse structure maximizes photosynthesis by using transparent materials to capture solar energy, increasing plant growth rates in poor soil conditions.**



You can only work on the next sections after you have built and tested your first prototype. You need to skip to the "Evidence is Reviewed" section.

STEM Principles for Design Solution #2

STEM Concept: The design solution is for the grass to grow at a consistent photosynthetic rate.



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Describe how STEM concept applies directly to the specifics of your solution:

STEM concepts apply to our solution because it is the blueprint for the steps to solve the problem to eventually get to a solution.

STEM Principles for Design Solution #3 (if completed)

STEM Concept: Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice.

Describe how STEM concept applies directly to the specifics of your solution:



Evidence Design is Reviewed (E3)

"Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

E3 is about experts reviewing what you've said about the STEM connections in E1 and E2. It's important here to talk to an appropriate field expert who knows about the STEM content of the project - not to a stakeholder.

Documentation #1 (Only one documentation section needed for full credit):

What format: Zoom call



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The expert stated that we came up with a surface-thought design that could maybe solve the solution. We were recommended to diagnose the problem, fully understand why the grass isn't growing, it could be the climate or unnoticed invasive species within the area. Encouraging the group to evaluate test results and prototype based on the brainstorms, tests, and feedback. Understanding the problem completely is an engineer's best friend when going about a solution process such as this one.

Documentation #2 (+ making only):

What format: In person

In our second interview we mainly went over some of the stuff we did last time but the main thing we were informed about how to obtain the soil kit and soil sample. She also Also informed us about other ways to improve the grass like using worms to help test soil/compost material and how it could promote a good grass area. She also told us about Stream health which is getting a net and collecting, counting different micro organisms to see how many there are. All of these insights were able to help us further think this through instead of just rushing to a solution which could have ruined our entire project.

Documentation #3 (+ making only):

What format: Interview notes, emails, audio/video tape with transcript, etc.

- What the expert said about the connection you've made from your design requirements to STEM principles and practices:
- What the expert said about how STEM principles and practices substantiate your design solution:



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Experts with Credentials (E4)

Document who your field experts are (names are not necessary) and what qualifies them as an expert. Use as many copies of this sub-template as needed.

Expert #1 (from documentation 1): Jeremy Diehlmann - Application Engineer at Victaulic

Expert #2 (from documentation 2): Describe credential(s) and what makes them qualified as an expert. Hannah Chop - Environmental Engineer

Expert #3 (from documentation 3): Describe credential(s) and what makes them qualified as an expert.

Verification or Corrective Response to Design Review (E5)

Your conversation/interview with the expert(s) should inform some corrective responses. In E5, describe how those reviews informed your responses.

For example, .

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will inform iterations you will make to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response: Mr. Deilman told us that we should slow down on the solution process to get to the design, which informed our decision to reconsider our design.

Interview with Expert #2 informs corrective response: Ms. Chop helped us further our project by informing us of a soil sample kit to help us know what is actually going on in the area instead of just rushing in with a solution without any prior knowledge.

Interview with Expert #3 informs corrective response:

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Criteria	5	4	3	2	1	0
3. Evidence that Design is Reviewed. There is _____ evidence that the application of STEM principles & practices (or suitable alternative) has been reviewed by expert(s).	<i>substantial</i>	<i>adequate</i>	<i>some</i>	<i>minimal</i>	<i>no - but review was requested -</i>	<i>no</i>
4. Design Reviewed by Experts. _____ expert(s) have/has reviewed the design for application of STEM principles & practices.	<i>Two or more, with at least two with documented, relevant credentials</i>	<i>Two or more, with at least one with documented, relevant credentials</i>	<i>At least one with documented, clearly relevant credentials</i>	<i>At least one with documented credentials</i>	<i>Only one with unclear credentials</i>	<i>No</i>
5. Design Review Results in Verification. The design reviews provided _____.	<i>clear and detailed confirmation (verification) or substantial detail necessary to inform a corrective response</i>	<i>clear confirmation (verification) or detail necessary to inform a corrective response</i>	<i>partial confirmation (verification) or some detail necessary to inform a corrective response</i>	<i>does not provide clear confirmation (verification) or suggestions for a corrective response, but at least some response is provided</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert,</i>	<i>show no evidence of confirmation (verification) or suggestions for a corrective response by an expert</i>



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		response				<i>but includes notes of attempts to gain feedback</i>	
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